Is Southeast Asia a Connector Economy in the Global Value Chain? Reflection from a Longitudinal Comparative Study, 2018–24

By
Ayadoure Srinivasane alias Stalin
UGC Senior Research Fellow, Centre for Indo Pacific Studies
School of International Studies, Jawaharlal Nehru University
New Delhi, India
ayadou57_isn@jnu.ac.in

Synopsis

1. Introduction
2. Methodology
3. Reviews
4. Results
5. Discussion
6. Limitations
7. Conclusions

INTRODUCTION

1. Changing economic relationships: China, the United States and India are important reference economies for examining Southeast Asia's evolving production connections.
2. Research problem: Increasing bilateral trade does not necessarily mean an economy has become a GVC connector; it may indicate greater dependence on a particular partner.
3. Research gap: The study examines whether changing intermediate-input linkages reflect intermediary capacity or increasing partner concentration.
4. Research objective: Compare changes in bilateral linkages, partner concentration and input-output-based intermediary relationships across the five economies.
5. Data and period: OECD Inter-Country Input-Output tables, with current empirical results covering 2018–2024.

DEFINITIONS

- **Global Value Chains (GVCs):** Production networks in which different stages of production, including: design, manufacturing, assembly and distribution, are located across multiple countries, with intermediate goods and services crossing borders for further production (OECD, 2013).
- **Connector Economy:** For this study, a connector economy is an economy that links production networks associated with multiple external partners through intermediate-input relationships. (Aiyar et al., 2026)
- **Intermediate Inputs:** Goods and services used as inputs in the production of other goods and services, rather than being purchased directly for final consumption or investment (OECD, 2013).
- **Bilateral Intermediate-Input Linkages:** Directed production relationships in which intermediate goods and services supplied by one economy are used by another economy's industries. Input-output tables map these inter-industry transactions within and across countries (OECD, 2013).
- **Partner Concentration:** The extent to which a country's measured linkages are concentrated among a limited number of external partners. The Herfindahl–Hirschman Index (HHI) measures concentration by summing the squared partner shares (U.S. Department of Justice, 2024).
- **Intermediary Capacity in GVCs:** The capacity of an economy to connect upstream suppliers and downstream production or demand through its intermediate-input relationships. GVC research distinguishes the use of foreign inputs in exports from the supply of domestic value added used in other economies' exports (OECD, 2013).

METHODOLOGY

1. **Research design:** Comparative longitudinal quantitative study of five Southeast Asian economies: Indonesia, Malaysia, the Philippines, Thailand and Viet Nam, covering 2018–2024.
2. **Data source:** OECD Inter-Country Input-Output (ICIO) year-wise data for 2018-24. 2025 not released yet.
3. **Objective**
    a. To aggregate industry-level intermediate transactions into directed country-to-country flows.
    b. To calculate partner shares and the Herfindahl–Hirschman Index (HHI) for China, the United States and India.
    c. To estimate direct-input pathway proxies to compare intermediary relationships across economies and destination markets.
    d. To compare 2018 and 2024 patterns and examine annual changes, subject to price-basis validation.

METHODOLOGY

Research Question:

- RQ1. How has Southeast Asia's role as a connector economy in global value chains (GVCs) changed between 2018 and 2025?
- RQ2. How have intermediate-input linkages between Indonesia, Malaysia, the Philippines, Thailand and Viet Nam, and China, the United States and India, changed over this period?
- RQ3. To what extent do these linkages indicate that Southeast Asian economies function as intermediaries within GVCs connecting China, the United States and India, rather than becoming increasingly dependent on a single partner?

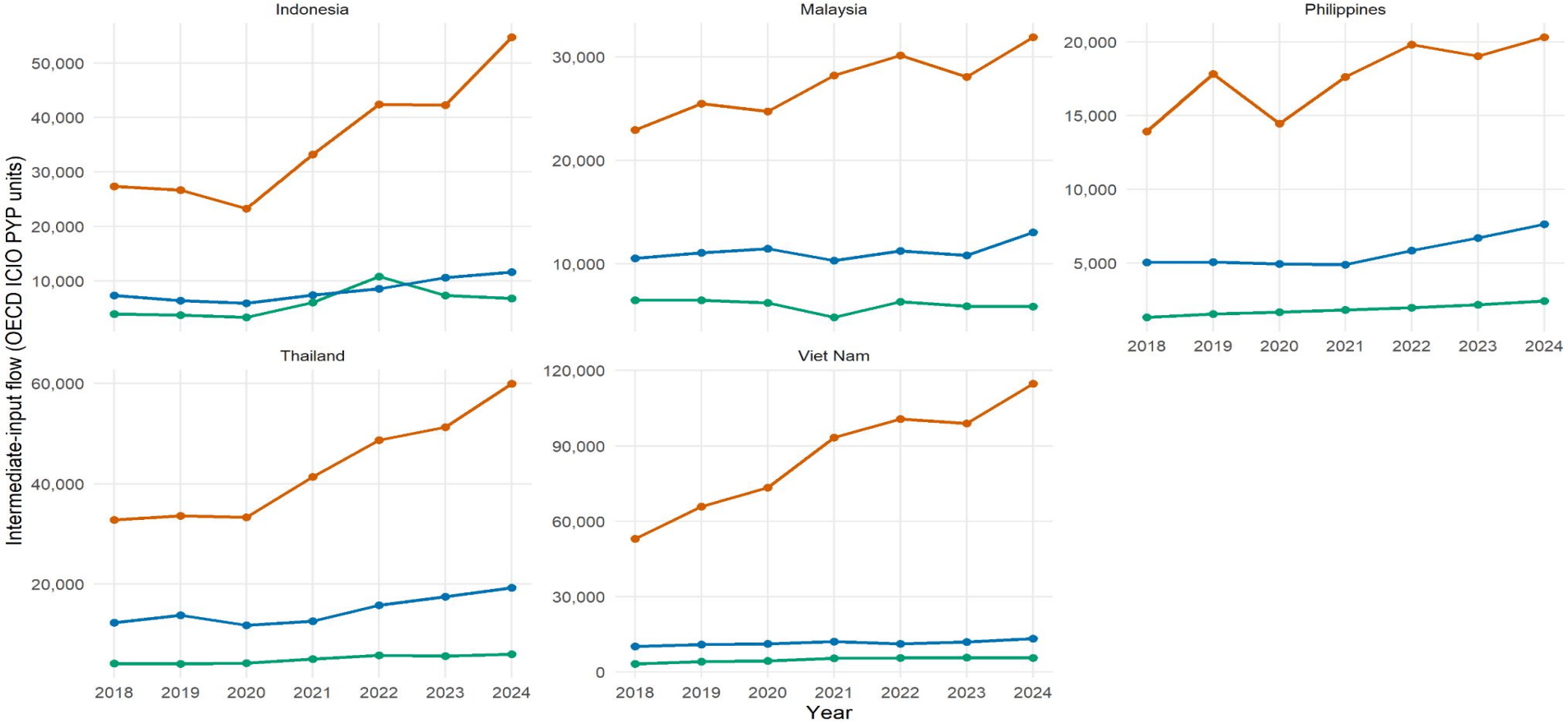
Research Hypotheses

- H1: Southeast Asian economies have experienced changes in the magnitude and composition of their intermediate-input linkages with China, the United States and India between 2018 and 2025, with the direction and extent of change varying across economies and partners.
- H2: Southeast Asian economies differ in their capacity to function as intermediaries within GVCs linking China, the United States and India, rather than merely experiencing an increase in bilateral dependence on one partner.

RESULT

China, the United States and India in Southeast Asia's Input Sourcing

Bilateral intermediate-input flows, 2018–2024



Source: OECD ICIO PYP dataset; author's calculations.

Table 2. Partner shares and Herfindahl–Hirschman Index (HHI), 2018–2024

Panel A. Annual partner shares and HHI

Economy	2018				2019				2020				2021				2022				2023				2024			
	China share (%)	US share (%)	India share (%)	HHI	China share (%)	US share (%)	India share (%)	HHI	China share (%)	US share (%)	India share (%)	HHI	China share (%)	US share (%)	India share (%)	HHI	China share (%)	US share (%)	India share (%)	HHI	China share (%)	US share (%)	India share (%)	HHI	China share (%)	US share (%)	India share (%)	HHI
Indonesia	62.84	17.06	20.10	0.4644	65.50	14.69	19.80	0.4899	68.43	14.84	16.73	0.5183	70.85	13.45	15.70	0.5447	67.06	12.47	20.47	0.5071	70.09	12.97	16.94	0.5368	72.00	12.92	15.07	0.5579
Malaysia	64.07	21.49	14.43	0.4776	66.26	21.09	12.65	0.4995	67.42	21.94	10.64	0.5140	69.52	20.66	9.82	0.5356	69.38	19.72	10.90	0.5321	70.83	18.88	10.30	0.5479	71.93	18.79	9.28	0.5613
Philippines	66.16	28.30	5.54	0.5209	67.97	26.64	5.39	0.5359	65.67	27.64	6.68	0.5122	69.35	24.24	6.42	0.5437	65.93	27.59	6.48	0.5150	63.61	29.13	7.26	0.4948	62.80	29.57	7.63	0.4876
Thailand	66.61	24.05	9.34	0.5103	65.78	25.79	8.43	0.5063	68.11	23.92	7.97	0.5275	69.05	22.36	8.60	0.5341	64.92	26.00	9.07	0.4974	65.04	26.52	8.44	0.5005	64.85	26.74	8.41	0.4992
Viet Nam	76.77	17.40	5.83	0.6231	76.14	18.27	5.59	0.6162	78.61	16.80	4.59	0.6483	78.65	16.58	4.77	0.6483	76.62	18.36	5.02	0.6233	77.58	17.22	5.20	0.6342	77.31	17.73	4.96	0.6316

Panel B. Summary of changes, 2018–2024

Economy	China share change (pp)	US share change (pp)	India share change (pp)	HHI 2018	HHI 2024	Δ HHI	Observed pattern
Indonesia	+9.16	−4.14	−5.03	0.4644	0.5579	+0.0935	**More concentrated**
Malaysia	+7.86	−2.70	−5.15	0.4776	0.5613	+0.0837	**More concentrated**
Philippines	−3.36	+1.27	+2.09	0.5209	0.4876	−0.0333	**More balanced**
Thailand	−1.76	+2.69	−0.93	0.5103	0.4992	−0.0111	**More balanced**
Viet Nam	+0.54	+0.33	−0.87	0.6231	0.6316	+0.0085	**Slightly more concentrated**

Notes: HHI is calculated as the sum of squared partner shares expressed as proportions. Higher HHI indicates greater concentration among China, the United States and India. Percentage-point (pp) changes are calculated from the displayed rounded shares, so minor rounding differences may occur.

Figure 3. Change in Connector-Linkage Pathway Intensity, 2018–2024

Direct-input pathway proxy

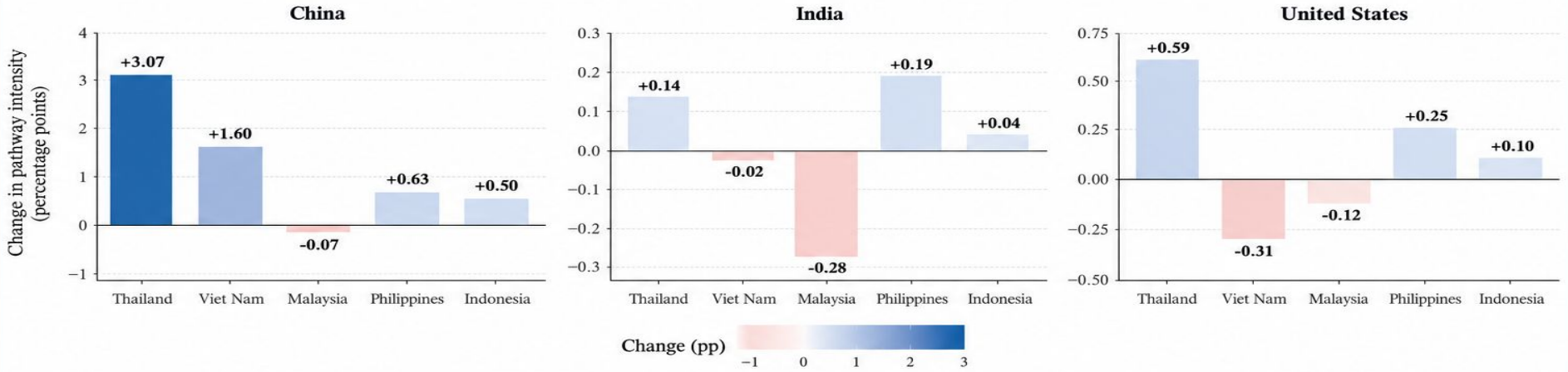


Table 3. Direct-Input Pathway Intensity, 2018 and 2024, and Change (percentage points)

Southeast Asian Economy	Origin Partner	Pathway Intensity 2018 (%)	Pathway Intensity 2024 (%)	Change (percentage points)
Indonesia	China	1.58	2.09	0.50
	United States	0.38	0.48	0.10
	India	0.22	0.26	0.04
Malaysia	China	3.96	3.90	-0.07
	United States	1.62	1.50	-0.12
	India	0.87	0.58	-0.28
Philippines	China	2.88	3.51	0.63
	United States	1.10	1.35	0.25
	India	0.22	0.42	0.19
Thailand	China	4.24	7.31	3.07
	United States	1.33	1.93	0.59
	India	0.44	0.58	0.14
Viet Nam	China	6.86	8.46	1.60
	United States	1.23	0.92	-0.31
	India	0.32	0.30	-0.02

DISCUSSION

RQ1. How has Southeast Asia's role as a connector economy in global value chains (GVCs) changed between 2018 and 2025?

- ➔ **China's persistent upstream dominance:** China-origin intermediate-input linkages remain substantially larger across Indonesia, Malaysia, the Philippines, Thailand and Viet Nam, reflecting its established position in regional production networks.
- ➔ **Declining US-origin intermediate inputs after 2022:** Examine the role of weaker global intermediate-goods trade, changes in industrial demand and sourcing patterns. These are potential explanations, not established causes.
- ➔ **Declining India-origin intermediate inputs after 2022:** Investigate whether sectoral specialisation and the composition of intermediate-input demand explain India's weakening linkages with the five Southeast Asian economies.
- ➔ **Geopolitical competition and RCEP:** US–China competition may reshape production locations, while RCEP facilitates regional integration among participating economies. Neither factor alone establishes the observed changes; China's dominance predates RCEP implementation.
- ➔ **GVC reconfiguration versus connector-economy intermediation:** The findings show persistent China-centred linkages and uneven changes across economies. Establishing stronger connector status requires tracing imported inputs into domestic production and exports to third markets.

RQ2. How have intermediate-input linkages between Indonesia, Malaysia, the Philippines, Thailand and Viet Nam, and China, the United States and India, changed over this period?

1. **China's relative position strengthened in Indonesia and Malaysia:** China's partner share rose by 9.16 and 7.86 percentage points, respectively, accompanied by higher HHI, indicating greater concentration among the three selected suppliers.
2. **The Philippines and Thailand became relatively more balanced:** The Philippines' China share declined by 3.36 percentage points, while its US and India shares increased. Thailand's China share declined by 1.76 points, alongside a 2.69-point increase in the US share.
3. **Viet Nam retained the highest supplier concentration:** China's share increased from 76.77% to 77.31%, while HHI rose from 0.6231 to 0.6316, indicating persistent concentration towards China.
4. **India's relative position varied across economies:** Its share declined in Indonesia, Malaysia, Thailand and Viet Nam, but increased in the Philippines. India's non-participation in RCEP provides institutional context, but does not independently explain these changes.
5. **Geopolitical competition and regional integration shape the context:** US–China strategic competition may encourage production-location diversification without displacing Chinese upstream inputs, while RCEP facilitates regional integration among participating economies. The observed patterns are consistent with these mechanisms but do not establish causation.

RQ3. To what extent do these linkages indicate that Southeast Asian economies function as intermediaries within GVCs connecting China, the United States and India, rather than becoming increasingly dependent on a single partner?

Thailand: Strongest increase in China-origin pathway intensity (+3.07 pp), with US and India pathways also rising.

Viet Nam: China-origin pathways increased (+1.60 pp), while US and India pathways declined.

Philippines & Indonesia: Pathway intensity increased across all three partners, led by China.

Malaysia: Pathway intensity declined across all three partners, particularly India (−0.28 pp).

Geopolitical interpretation: China-centred production networks remain influential, while US–China strategic competition and manufacturing diversification shape the wider context of regional production.

Overall finding: Southeast Asia exhibits uneven, asymmetrical reconfiguration of GVC linkages—not uniform diversification or convergence towards a single partner.

H1 Robustness Results

Changes in intermediate-input linkages with China, United States and India (2018–2024)

Table 1. Endpoint comparison of bilateral intermediate-input flows

Southeast Asian Economy	Partner	Flow 2018	Flow 2022	Flow 2024	Change 2018–2022	Change 2018–2024
Indonesia	China	27,375	42,382	54,758	15,008	27,384
	India	3,885	10,795	6,759	6,909	2,874
	United States	7,280	8,524	11,605	1,245	4,325
Malaysia	China	22,931	30,123	31,893	7,192	8,962
	India	6,499	6,343	5,897	−156	−602
	United States	10,539	11,268	13,043	730	2,504
Philippines	China	13,923	19,819	20,326	5,897	6,404
	India	1,305	1,972	2,428	668	1,123
	United States	5,031	5,858	7,646	827	2,615
Thailand	China	32,782	48,677	59,954	15,896	27,172
	India	4,172	5,813	6,030	1,641	1,858
	United States	12,302	15,826	19,311	3,523	7,008
Viet Nam	China	52,959	100,778	114,784	47,820	61,825
	India	3,196	5,691	5,617	2,495	2,421
	United States	10,205	11,257	13,270	1,052	3,065

Table 2. Direction of change: robustness check (2018–2022 vs 2018–2024)

Comparisons (economy–partner pairs)	Same direction of change	Opposite direction of change
15	15	0

Table 3. Partner-wise summary of changes (2018–2024)

Partner	Increased (of 5)	Decreased (of 5)	Key observations
China	5	0	Increased for all five economies (largest increase in Viet Nam)
India	4	1	Increased for four economies; decreased for Malaysia
United States	5	0	Increased for all five economies (largest increase in Thailand)

H1 Assessment

Accepted (descriptive support)

- All 15 economy–partner pairs show the same direction of change for 2018–2022 and 2018–2024.
- China and United States flows increased for all five economies.
- India flows increased for four economies and decreased for Malaysia.
- Robust to endpoint comparison; however, statistical significance and causal interpretation are not yet established.

H2 Robustness Results

GVC intermediation between China, United States and India (2018–2024)

Table 1. Change in connector-linkage pathway values (2018–2024)

Intermediary economy	Total pathways	Increased	Decreased	Unchanged	Total 2018	Total 2024	Net change
Indonesia	6	6	0	0	1,987	3,926	1,939
Malaysia	6	5	1	0	5,608	6,910	1,301
Philippines	6	6	0	0	1,335	2,068	733
Thailand	6	6	0	0	5,435	13,191	7,756
Viet Nam	6	6	0	0	9,823	27,110	17,287

Table 2. Relative pathway growth by intermediary economy

Intermediary economy	Pathways increased	Pathways decreased	Median growth (%)	Mean growth (%)	Median absolute change
Thailand	6	0	103.0	106.0	881.6
Viet Nam	6	0	97.8	101.0	757.3
Indonesia	6	0	91.6	104.0	351.1
Philippines	6	0	74.5	80.0	267.3
Malaysia	5	1	9.65	15.0	95.9

Table 3. Destination-wise net change in pathway values (2018–2024)

Intermediary economy	Destination China	Destination India	Destination United States	Total net change
Indonesia	519	382	1,037	1,939
Malaysia	737	78.7	486	1,301
Philippines	163	96.5	473	733
Thailand	812	940	6,004	7,756
Viet Nam	807	774	15,706	17,287

Table 4. Pathway coverage

Intermediary economy	Total pathways	Cross-partner pathways (origin ≠ destination)
Indonesia	6	6
Malaysia	6	6
Philippines	6	6
Thailand	6	6
Viet Nam	6	6

H2 Assessment

Partial support (descriptive evidence)

- 29 of 30 pathways show an increase between 2018 and 2024.
- Thailand and Viet Nam show the largest pathway expansion; Malaysia shows the weakest growth.
- Positive changes for all destinations (China, India and United States), though US-destination pathways are particularly large for Thailand and Viet Nam.
- Provides descriptive support for differences in intermediary capacity, but requires further validation (normalisation, pre-projection period, and statistical testing).

LIMITATION

1. **Temporal coverage:** The analysis covers 2018–2024 rather than the full hypothesised period of 2018–2025. The 2023–2024 ICIO PYP observations are projected, requiring caution when interpreting recent changes.
2. **Limited partner selection:** The study focuses on China, the United States and India. The findings therefore capture changes within these selected relationships, not the entirety of Southeast Asia's global value chain (GVC) linkages.
3. **Measurement of intermediate-input flows:** Bilateral intermediate-input flows measure the magnitude of transactions, while supplier shares and the three-partner Herfindahl–Hirschman Index (HHI) capture relative composition and concentration. These indicators do not independently measure overall GVC dependence or domestic value added.
4. **Operationalisation of connector-economy status:** The direct-input pathway proxy provides an indication of changing production linkages but does not, on its own, establish that inputs originating in one partner are transformed within a Southeast Asian economy and subsequently embodied in exports to another partner. The origin–intermediary–destination measures require validation against the underlying input-output structure.
5. **Limited robustness and statistical inference:** The endpoint comparison shows consistent directions of change across 15 economy–partner pairs, while the pathway analysis records increases in 29 of 30 constructed routes. These are descriptive robustness findings, not independent statistical significance tests. The short time series also limits the scope for reliable statistical inference.
6. **Causal attribution:** The study situates observed changes within contemporary geopolitical developments, including US–China strategic competition, manufacturing diversification and regional economic integration. However, the descriptive ICIO analysis cannot establish that these developments caused the observed changes in intermediate-input linkages.
7. **Aggregation across industries:** Aggregate country-level flows may conceal substantial differences across sectors, including electronics, machinery, chemicals and other manufacturing industries. Without further sector-level decomposition, it is difficult to identify which industries drive the observed changes.
8. **Comparability of pathway measures:** Absolute pathway values and percentage changes produce different country rankings. Comparisons of intermediary capacity therefore require appropriate normalisation by export or production scale and consistent pathway definitions.

CONCLUSION

- **Uneven reconfiguration of GVC linkages:** Between 2018 and 2024, intermediate-input linkages between Indonesia, Malaysia, the Philippines, Thailand and Viet Nam and China, the United States and India changed in magnitude and composition, with different trajectories across economies and partners.
- **China's continued importance:** China-origin intermediate-input flows increased across all five economies. However, changes in supplier concentration varied, indicating that stronger bilateral linkages did not translate uniformly into greater concentration towards China.
- **Differentiated intermediary pathways:** The constructed pathway measures increased across 29 of 30 cross-partner routes. Thailand and Viet Nam recorded the largest increases in summed pathway values, while Malaysia experienced the weakest expansion.
- **Partial evidence for the connector-economy argument:** The findings indicate expanding cross-partner production pathways and differences across Southeast Asian economies. Nevertheless, pathway growth alone does not conclusively establish that inputs from one partner are embodied in exports to another.
- **Contribution and future research:** The study demonstrates the value of combining bilateral flows, supplier composition and pathway measures to examine Southeast Asia's evolving GVC position. Further research should validate origin–intermediary–destination pathways, examine sector-level differences, test sensitivity to projected observations and assess how geopolitical developments relate to these changes.

REFERENCE (APA FORMAT)

1. Antràs, P. (2020). Conceptual aspects of global value chains. The World Bank Economic Review, 34(3), 551–574. https://doi.org/10.1093/wber/lhaa006

2. Asian Development Bank. (2024). Asian economic integration report 2024: Decarbonizing global value chains. Asian Development Bank. https://doi.org/10.22617/SGP240085-2

3. Baldwin, R. (2011). 21st century regionalism: Filling the gap between 21st century trade and 20th century trade rules (WTO Staff Working Paper No. ERSD-2011-08). World Trade Organization. https://doi.org/10.30875/c67646a3-en

4. Baldwin, R. (2016). The great convergence: Information technology and the new globalization. Harvard University Press.

5. Baldwin, R., & Freeman, R. (2022). Risks and global supply chains: What we know and what we need to know. Annual Review of Economics, 14, 153–180. https://doi.org/10.1146/annurev-economics-051420-113737

6. Gereffi, G., Humphrey, J., & Sturgeon, T. (2005). The governance of global value chains. Review of International Political Economy, 12(1), 78–104. https://doi.org/10.1080/09692290500049805

7. Hummels, D., Ishii, J., & Yi, K.-M. (2001). The nature and growth of vertical specialization in world trade. Journal of International Economics, 54(1), 75–96. https://doi.org/10.1016/S0022-1996(00)00093-3

8. Humphrey, J., & Schmitz, H. (2002). How does insertion in global value chains affect upgrading in industrial clusters? Regional Studies, 36(9), 1017–1027. https://doi.org/10.1080/0034340022000022198

9. Johnson, R. C., & Noguera, G. (2012). Accounting for intermediates: Production sharing and trade in value added. Journal of International Economics, 86(2), 224–236. https://doi.org/10.1016/j.jinteco.2011.10.003

10. Koopman, R., Wang, Z., & Wei, S.-J. (2014). Tracing value-added and double counting in gross exports. American Economic Review, 104(2), 459–494. https://doi.org/10.1257/aer.104.2.459

11. Organisation for Economic Co-operation and Development. (2025). Inter-country input-output tables (ICIO) [Data set]. https://www.oecd.org/en/data/datasets/inter-country-input-output-tables.html

12. Timmer, M. P., Erumban, A. A., Los, B., Stehrer, R., & de Vries, G. J. (2014). Slicing up global value chains. Journal of Economic Perspectives, 28(2), 99–118. https://doi.org/10.1257/jep.28.2.99

13. United Nations Conference on Trade and Development. (2024). World investment report 2024: Investment facilitation and digital government. United Nations. https://unctad.org/system/files/official-document/wir2024_en.pdf

14. World Trade Organization, Institute of Developing Economies–Japan External Trade Organization, Organisation for Economic Co-operation and Development, Research Institute of Global Value Chains at the University of International Business and Economics, & World Bank. (2023). Global value chain development report 2023: Resilient and sustainable GVCs in turbulent times. World Trade Organization.

15. World Trade Organization. (2023). World trade report 2023: Re-globalization for a secure, inclusive and sustainable future. World Trade Organization. https://www.wto.org/english/res_e/booksp_e/wtr23_e/wtr23_e.pdf

THANK YOU